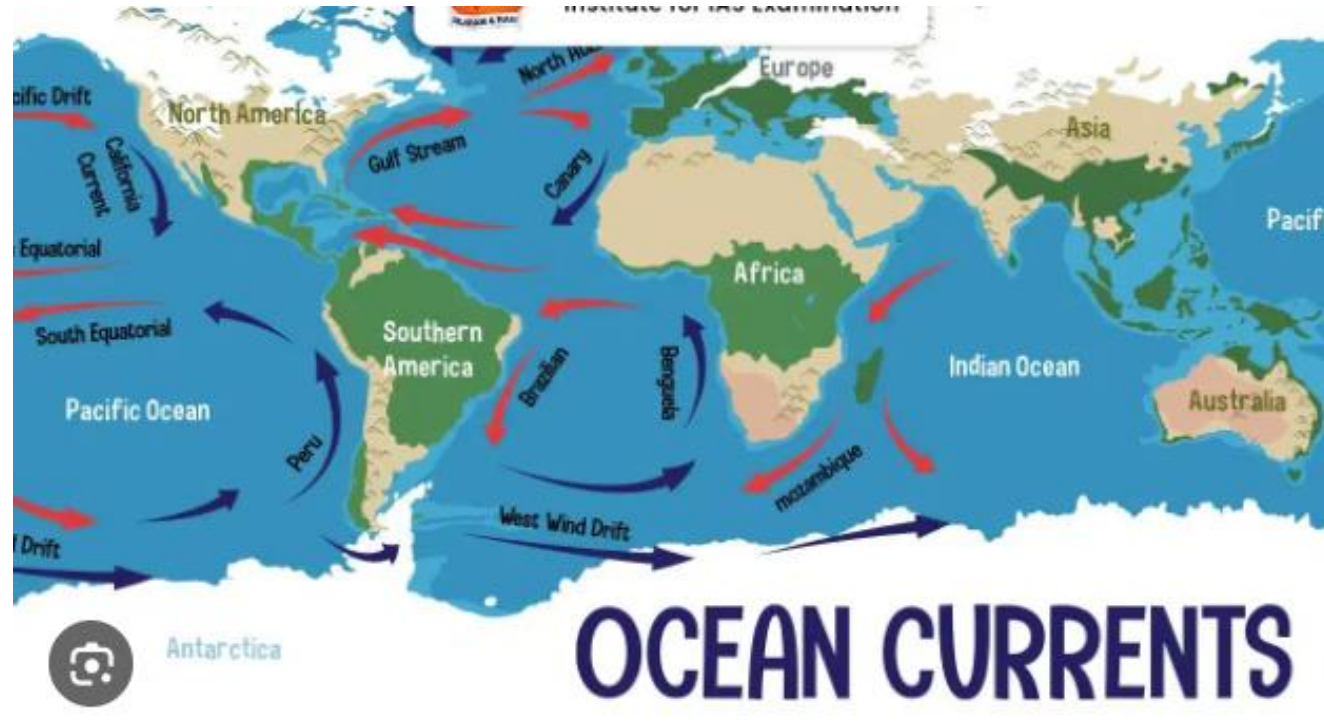
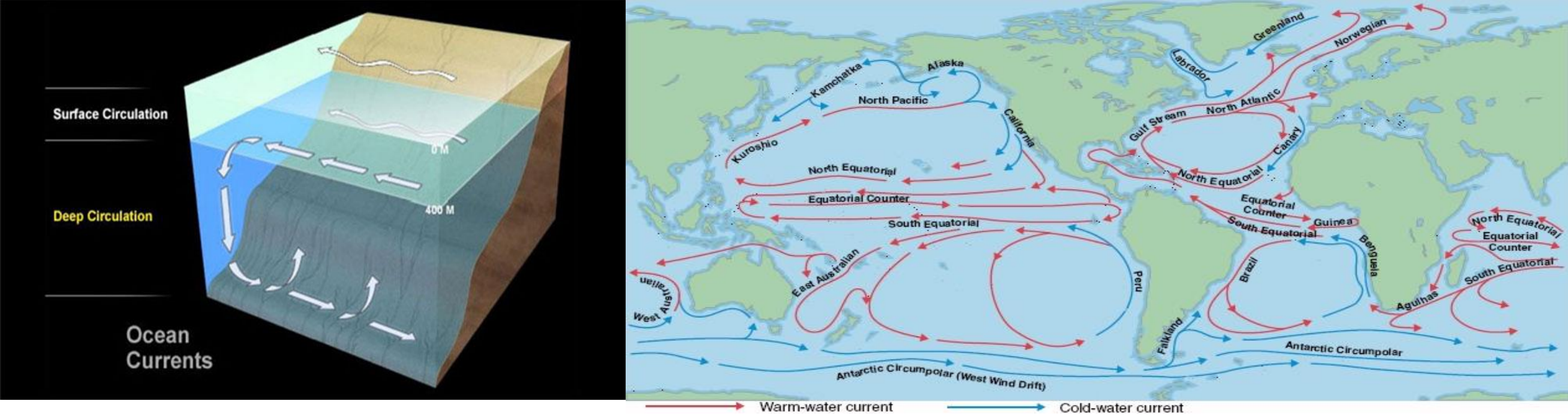


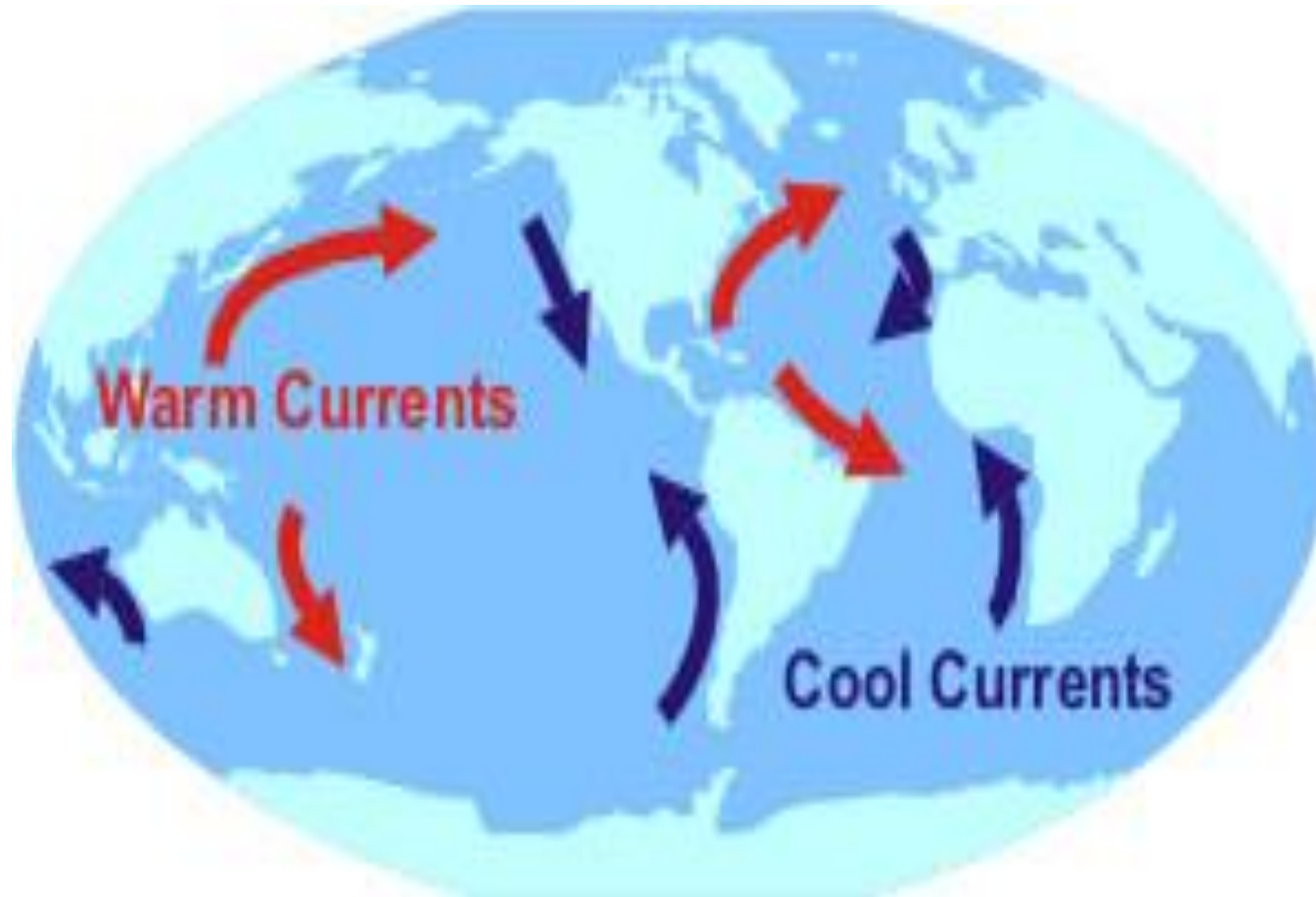
OCEAN CURRENTS





- **Cold currents** are those which flow towards the equator and move towards the tropics or lower latitudes which are on the eastern side of ocean basins.
- They bring cold water into those areas where there is warm water which are mainly observed in the west coasts of the continent.
- Ocean water moves in two directions: horizontally and vertically.
 - Horizontal movements are referred to as currents, while vertical changes are called upwellings or downwellings.

COLD CURRENTS



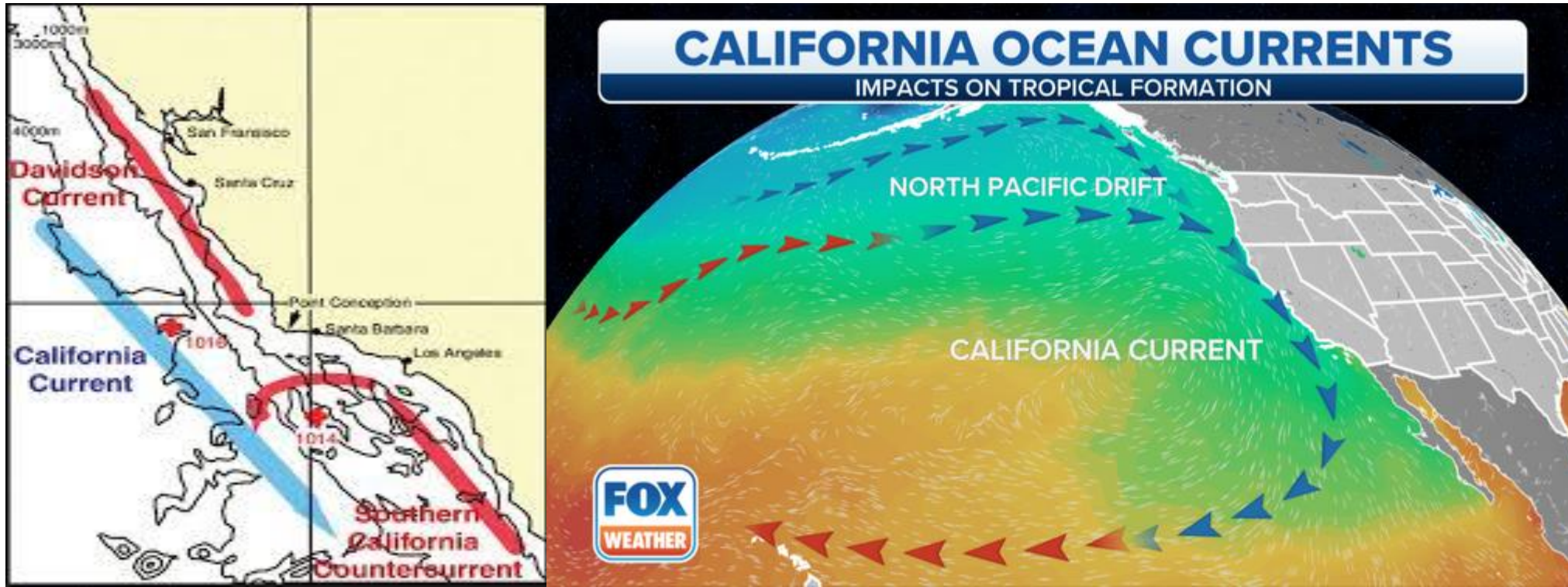
HUMBOLDT OR PERUVIAN COLD CURRENT

The Humboldt Current, or Peru Current, is a cold, low-salinity ocean current flowing northward along the western coast of South America (Chile, Peru, Ecuador). Driven by southeast trade winds, it brings cold, nutrient-rich water from the Antarctic to the surface (upwelling), creating one of the world's most productive marine ecosystems and supporting major fisheries.



CALIFORNIA COLD CURRENT

The California Current is a cold, southward-flowing Pacific Ocean current (part of the North Pacific Gyre) traveling along the western coast of North America from British Columbia to Baja California. It brings cold, nutrient-rich water from Alaska southward, causing significant summer coastal fog, creating a cooler, arid climate, and supporting rich marine ecosystems.



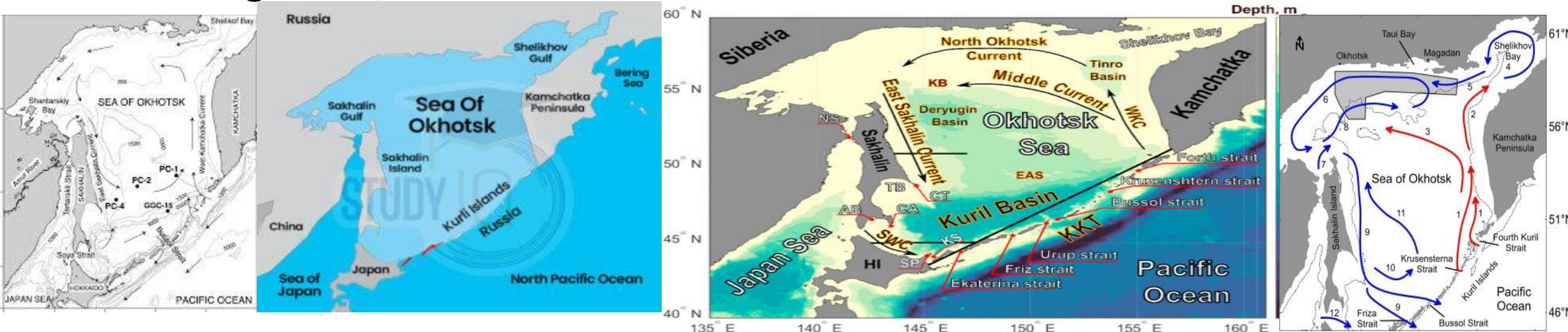
ANTARTICA CIRCUMPOLAR COLD CURRENT



- Also known as the **West Wind Drift**
- **Strongest** and **largest** wind-driven current on Earth
- Moves **clockwise** around **Antarctica** when viewed from the **South Pole**
- Only current in the global ocean to close upon itself in a **circumpolar loop**
- **Driven by strong westerly winds**
- Only current that goes all the way around the planet and connects the **Atlantic, Pacific, and Indian Oceans**
- **Massive flow** of water acting as a barrier, separating the **Southern Ocean** from more northern oceans

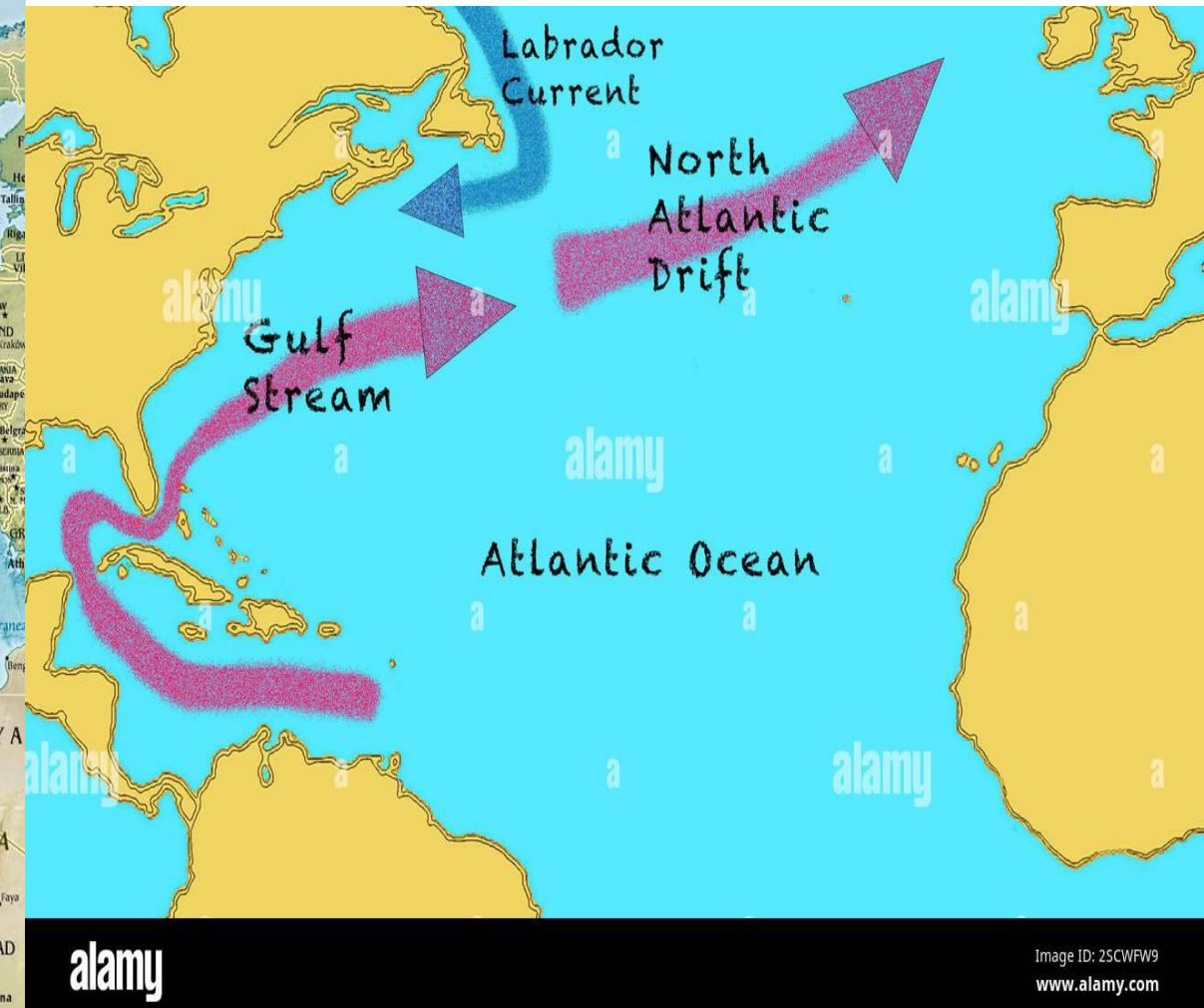
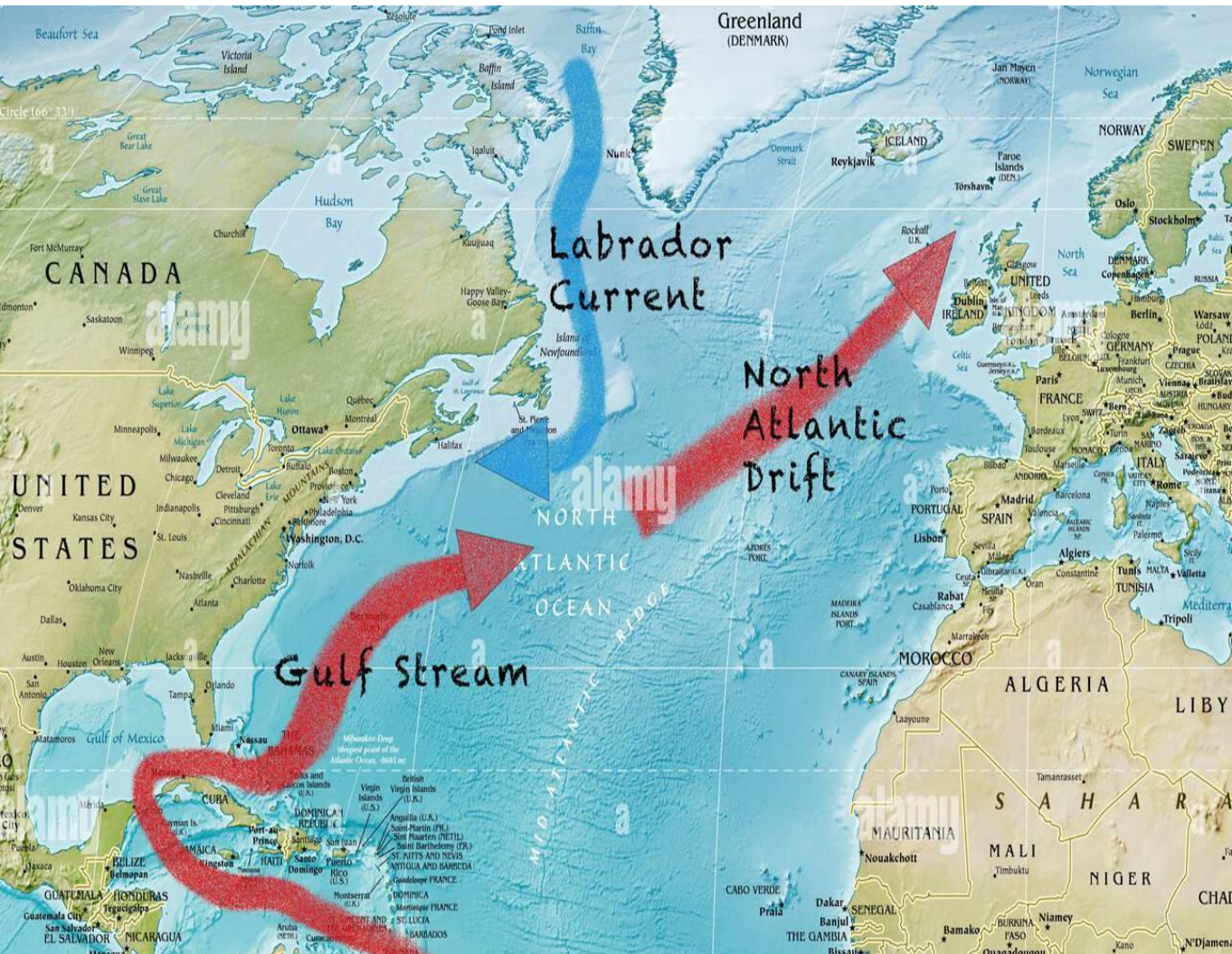
The Oyashio Current "Parental Tide"), also known as the Okhotsk Current or Kurile Current

- The **Okhotsk Cold Current**, is a frigid, subarctic ocean current that flows southward and circulates counterclockwise in the western North Pacific Ocean.
- It is considered the coldest sea in East Asia, playing a crucial role in regulating climate, driving marine productivity, and controlling sea ice distribution in the region.
- The current originates in the Arctic Ocean, flows south through the Bering Sea and Bering Strait, and enters the Sea of Okhotsk.



LABRADOR COLD CURRENT

**surface oceanic current flowing southward along the west side of the Labrador Sea.
Flowing through the Davis Strait, the Labrador Current is a combination of the West Greenland Current, the Baffin Island Current, and inflow from Hudson Strait**



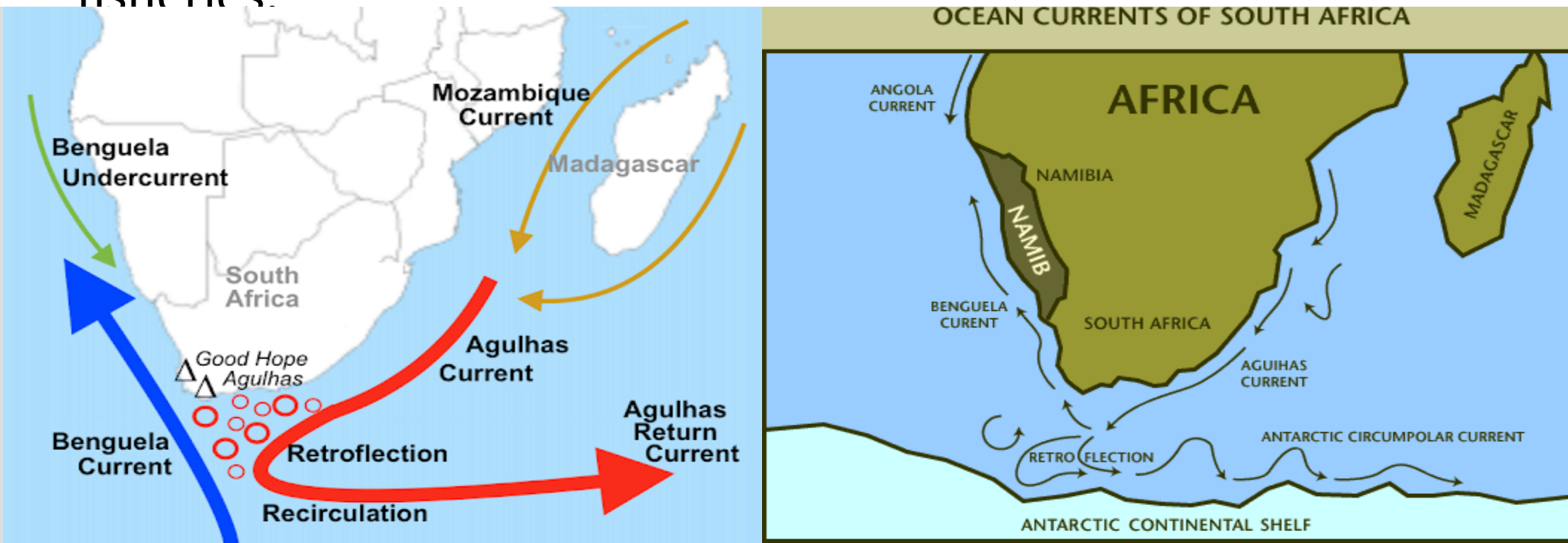
EASTERN GREENLAND COLD CURRENT

- The East Greenland Current (EGC) is a major cold, low-salinity ocean current flowing southward along the eastern coast of Greenland, acting as the primary conduit for Arctic Ocean water, sea ice, and fresh water to enter the North Atlantic.
- It originates in the Fram Strait, flows through the Denmark Strait, and is critical for global climate and oceanic circulation.



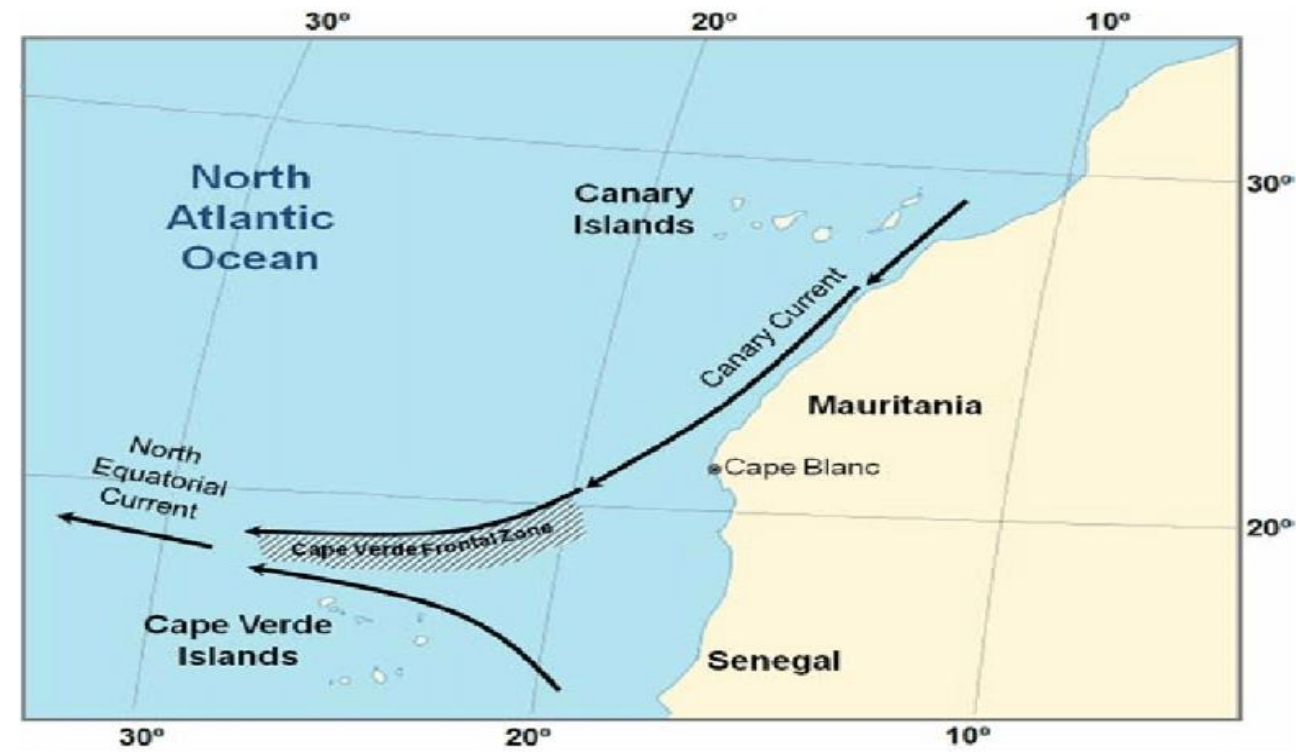
BEUGUELA COLD CURRENT

- The cold Benguela Current is a nutrient-rich, northward-flowing ocean current along the southwestern coast of Africa (South Africa, Namibia, Angola).
- Driven by southeast trade winds, it causes intense coastal upwelling of cold water, making it one of the world's most productive marine ecosystems.
- It creates cool, arid coastal conditions, impacting climate and sustaining rich fisheries.



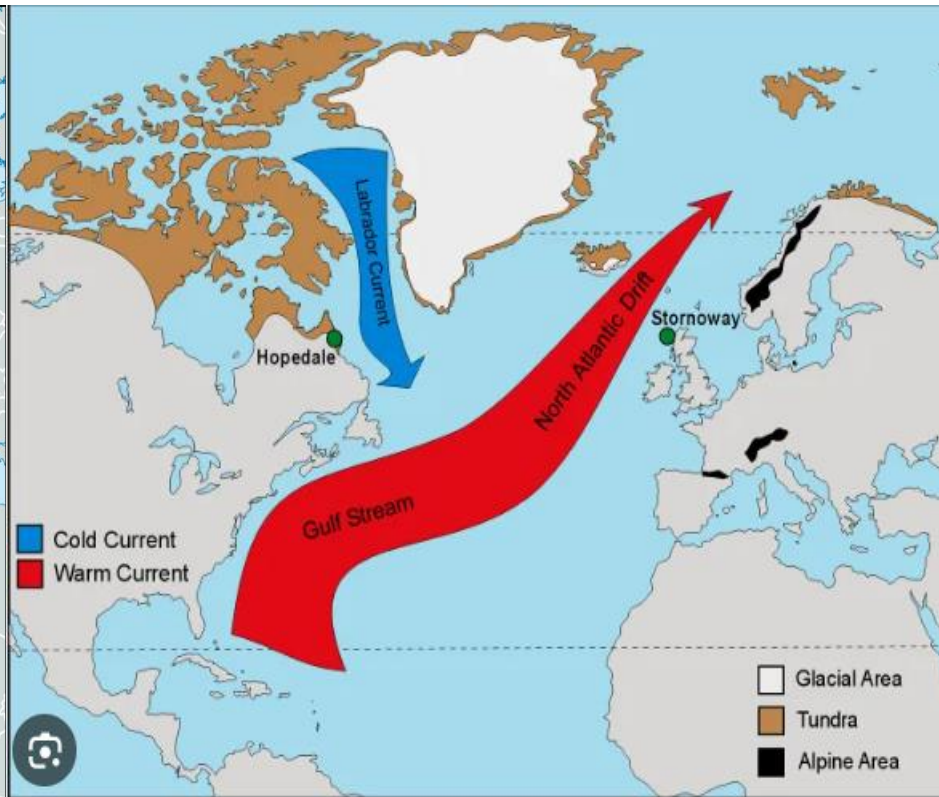
CANARY CURRENT

- The Canary Current is a cold, slow-moving wind-driven surface current in the North Atlantic Ocean that flows southwestward along the northwest coast of Africa. It is an Eastern Boundary Current of the North Atlantic Subtropical Gyre, originating near the Azores and bringing nutrient-rich waters that support significant marine life and fisheries. The current, which flows between Madeira and Cape Verde, is highly influenced by trade winds and causes upwelling



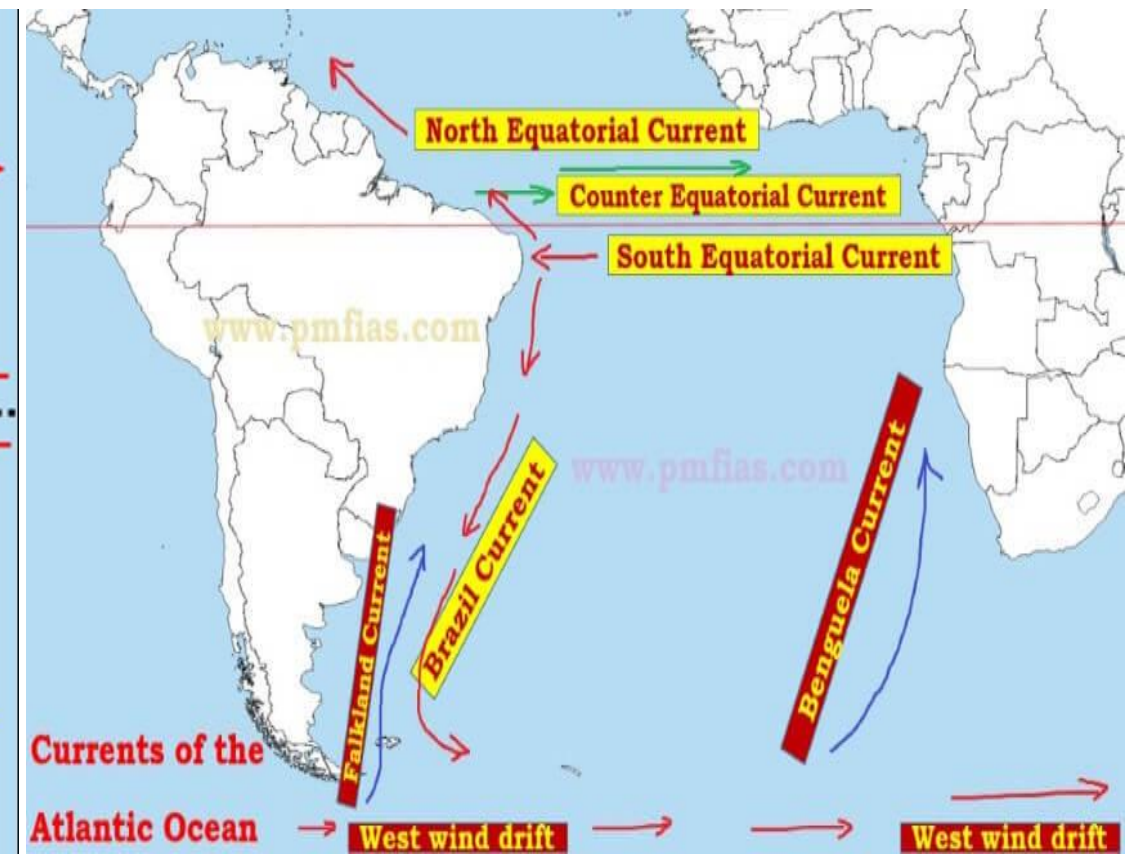
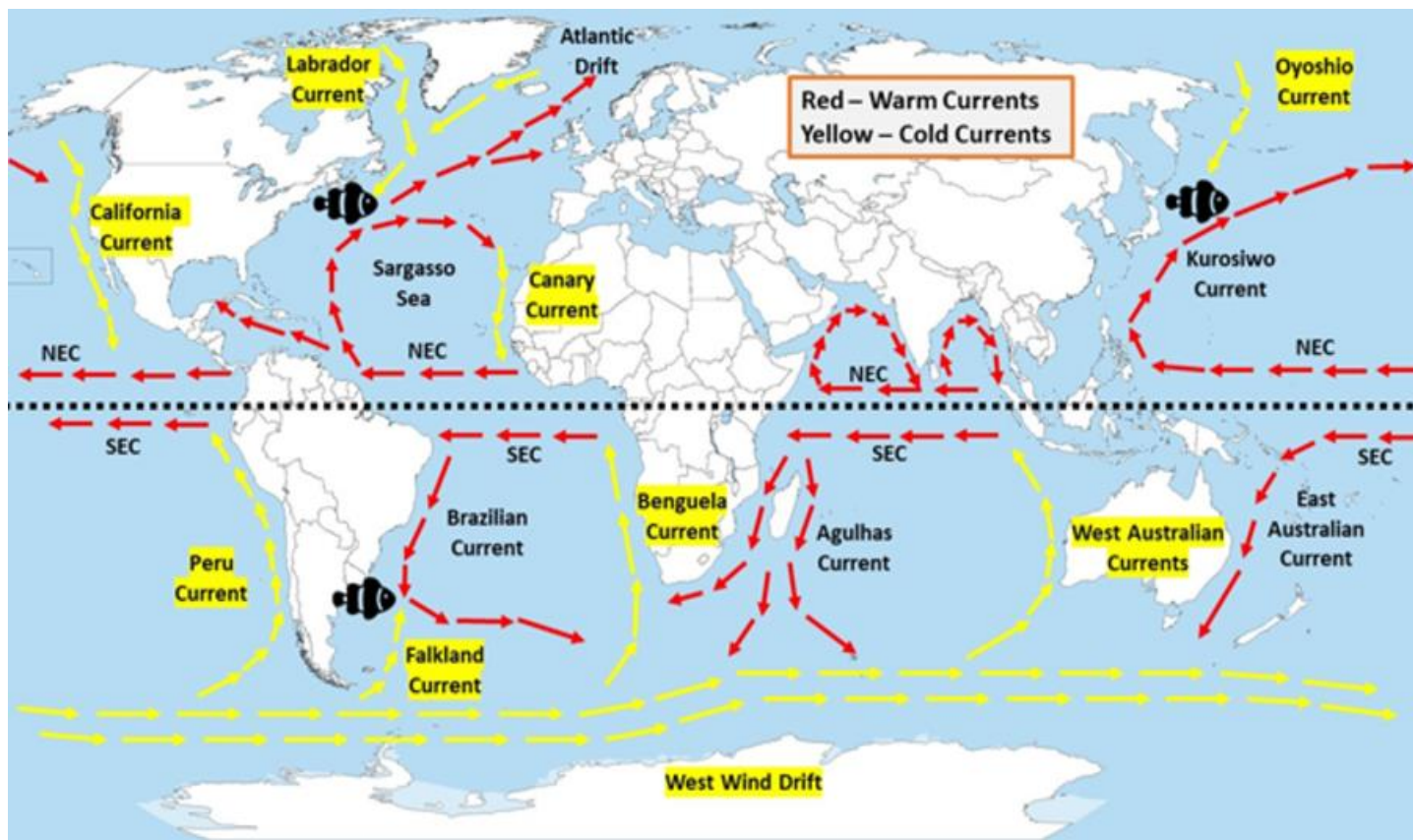
CANARY COLD CURRENT

- The Canary Current is a cold, wind-driven surface current in the North Atlantic Ocean that flows southward along the northwest coast of Africa (from Iberia to Senegal).
- As an eastern boundary current of the North Atlantic Gyre, it brings nutrient-rich, cool water (12-16°C) to the region, creating productive fishing grounds and aiding in the formation of the Sahara Desert by stabilizing the air.



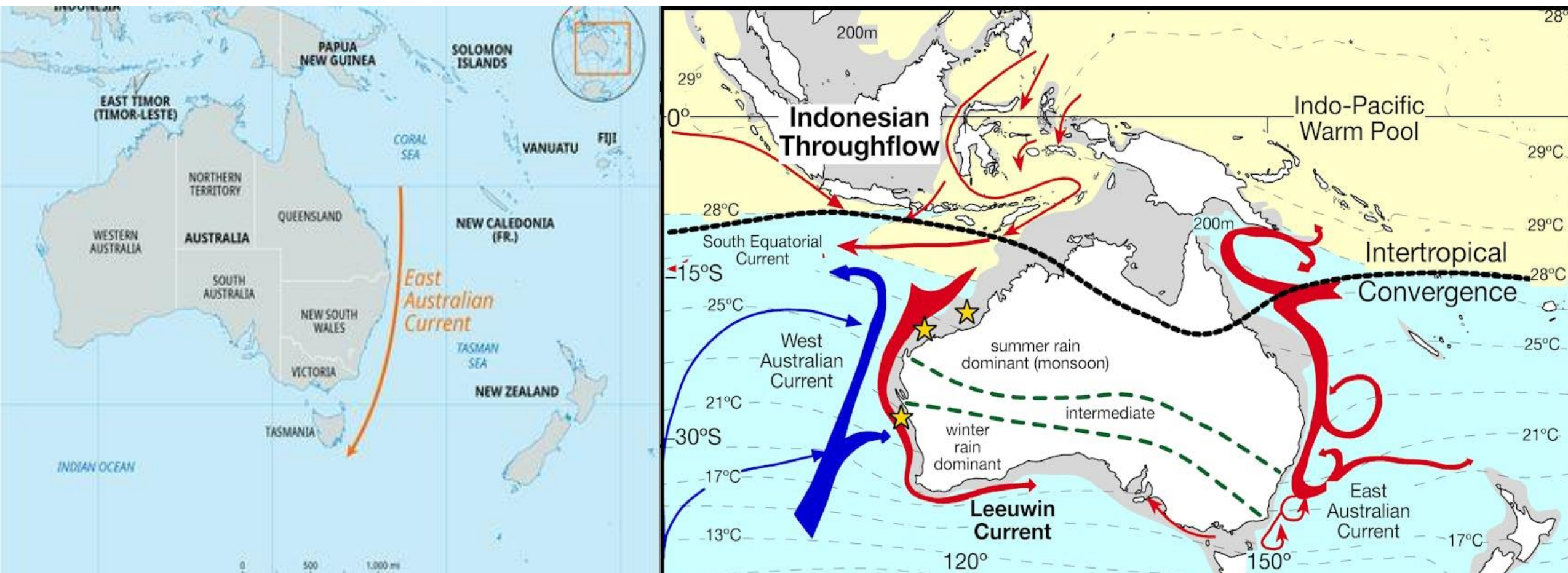
FALKLAND COLD CURRENT

- The Falkland Current (or Malvinas Current) is a cold, nutrient-rich ocean current flowing northward along the Atlantic coast of Patagonia to the mouth of the Río de la Plata.
- It flows along the east coast of Argentina, generally parallel to the coast.



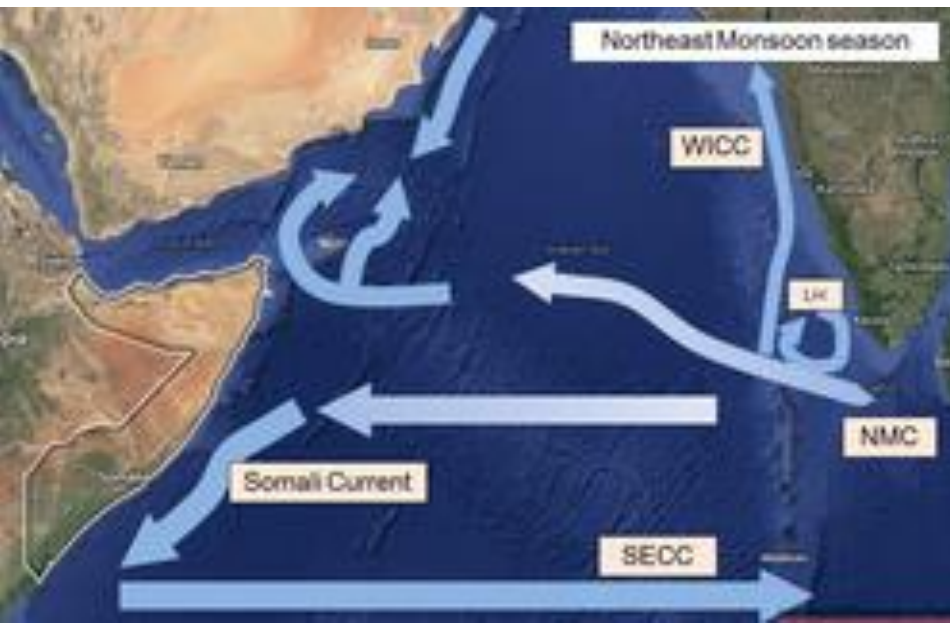
West Australian Cold Current

- The West Australian Current (WAC) is a cold, northward-flowing ocean current in the southeast Indian Ocean that acts as part of the Antarctic Circumpolar Current system. It is a seasonal, wind-driven, and nutrient-rich current that is generally stronger in summer. It flows along the western coast of Australia,



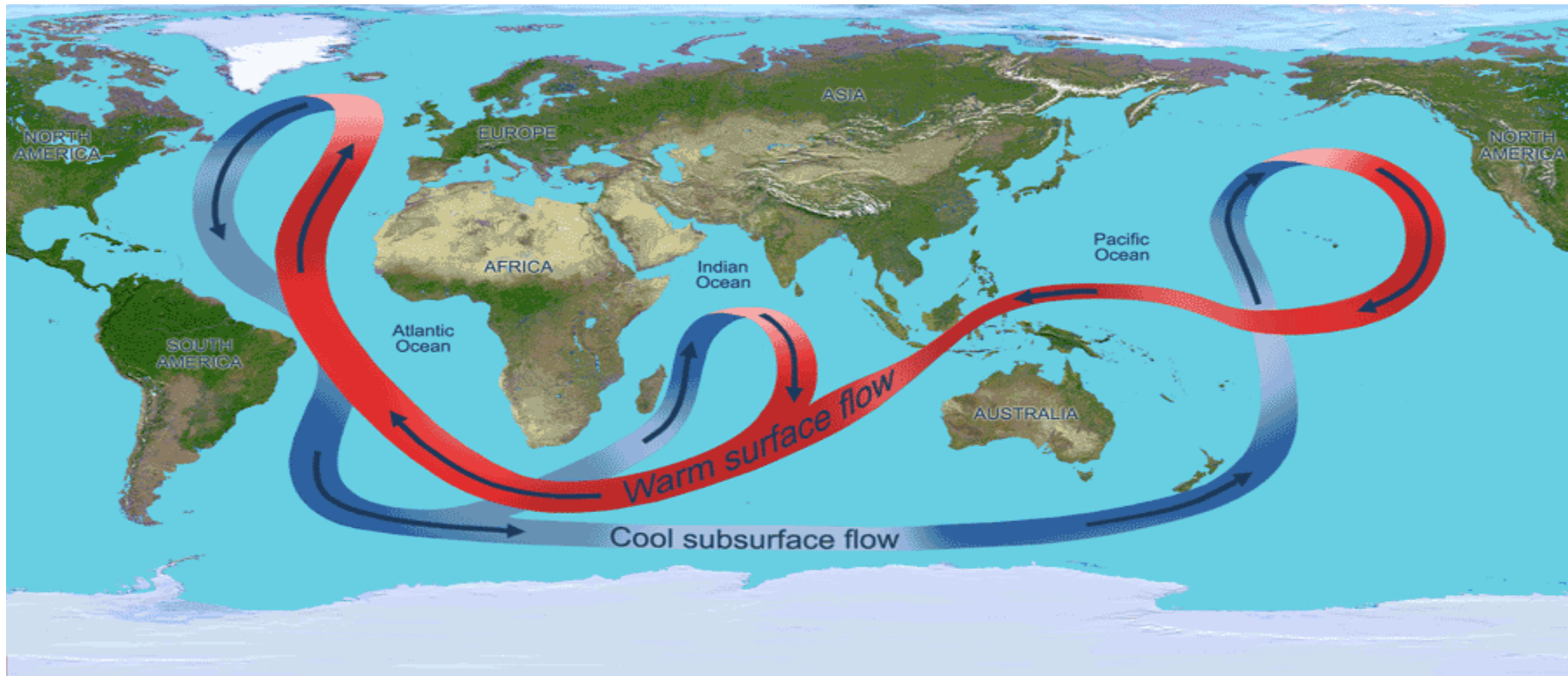
Somali Cold Current

- The Somali Current is a cold ocean boundary current that runs along the coast of Somalia and Oman in the Western Indian Ocean and is analogous to the Gulf Stream in the Atlantic Ocean.
- Analogous to the Gulf Stream in the Atlantic Ocean.
The Current is heavily influenced by monsoon.
 - Region of major upwelling system.



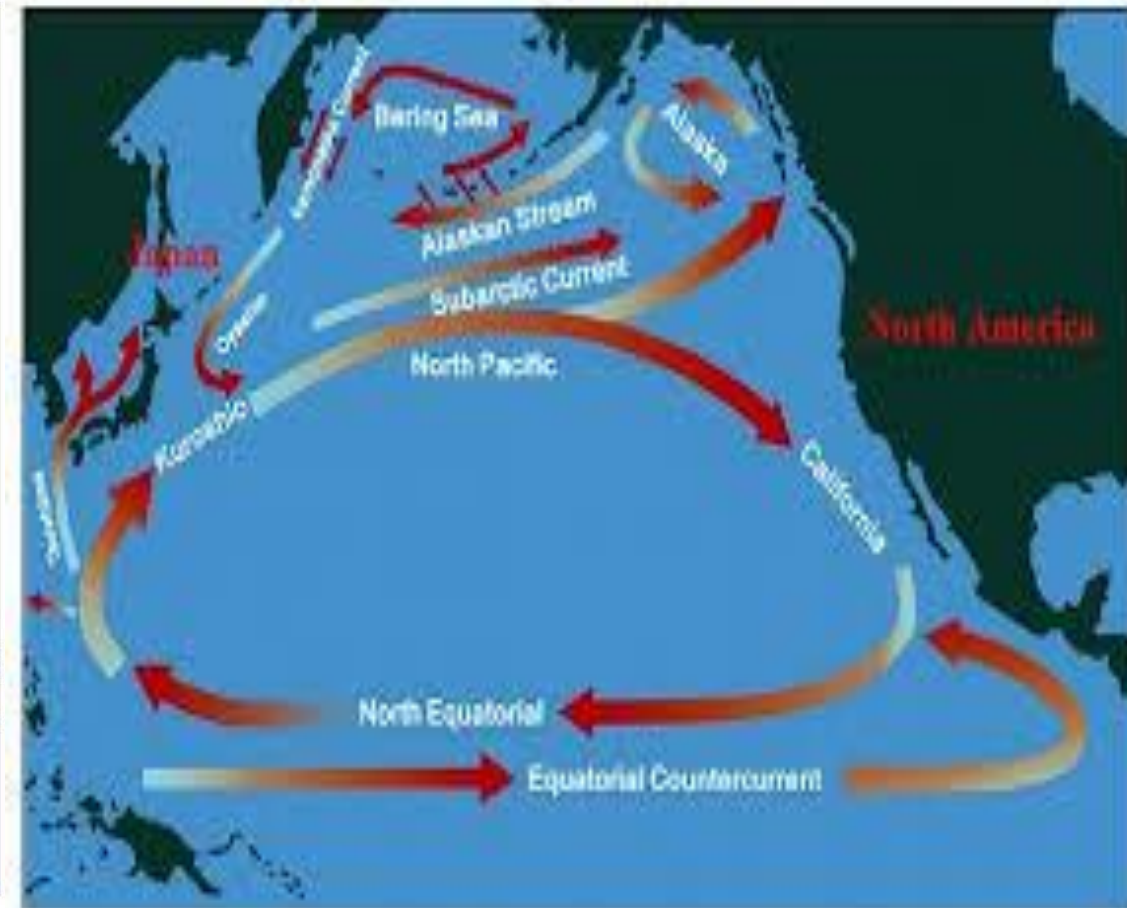
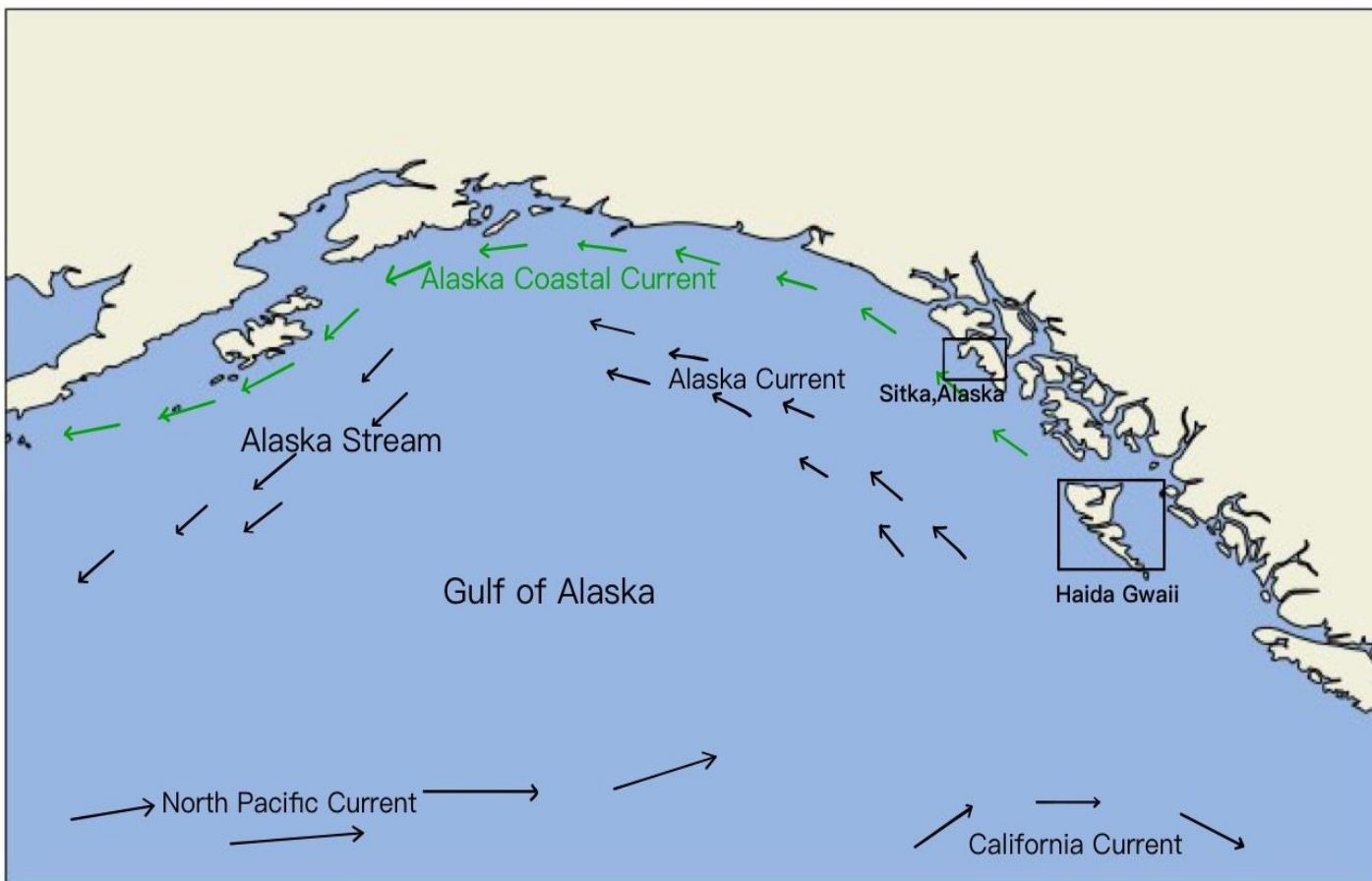
WARM CURRENTS

- Warm currents are those that move from the Equator to the poles and are warmer than the surrounding water.
- In the low and medium altitudes, these currents are frequently seen on the east coasts of continents (true in both hemispheres).



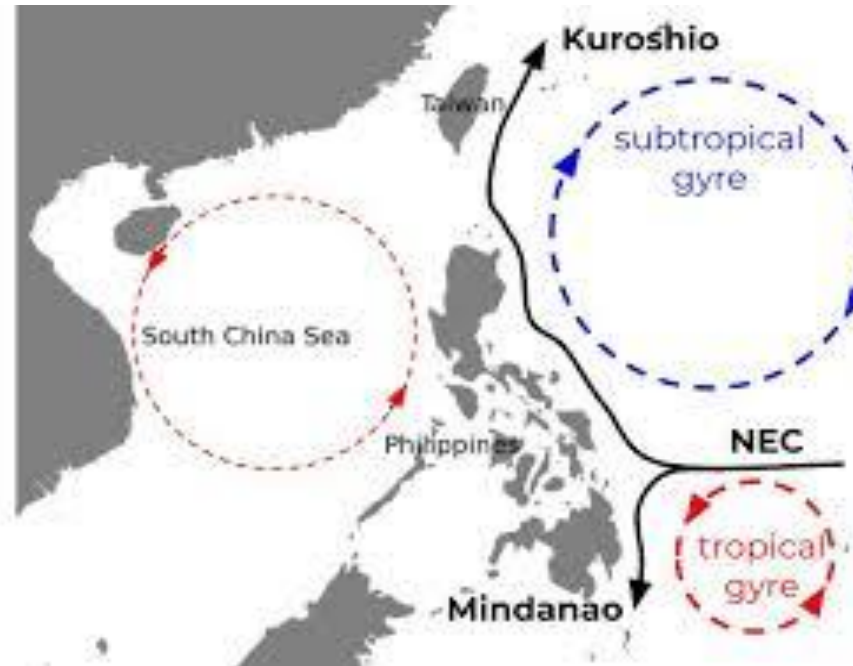
ALASKAN WARM CURRENT

- The Alaska Current is a southwestern shallow warm-water current alongside the west coast of the North American continent beginning at about 48-50°N. The Alaska Current produces large clockwise eddies at two sites: west of Haida Gwaii and west of Sitka, Alaska



Kuroshio warm Current

- This west boundary current is also known as Japan current or Black Current. The term **“Kuroshio” in Japanese means “Black Stream”**.
- The Kuroshio Current ("Black Tide"), also known as the Black Current or Japan Current (Nihon Kairyū), is a north-flowing, warm ocean current on the west side of the North Pacific Ocean basin.
- It was named for the deep blue appearance of its waters.



Northeast Monsoon warm Current

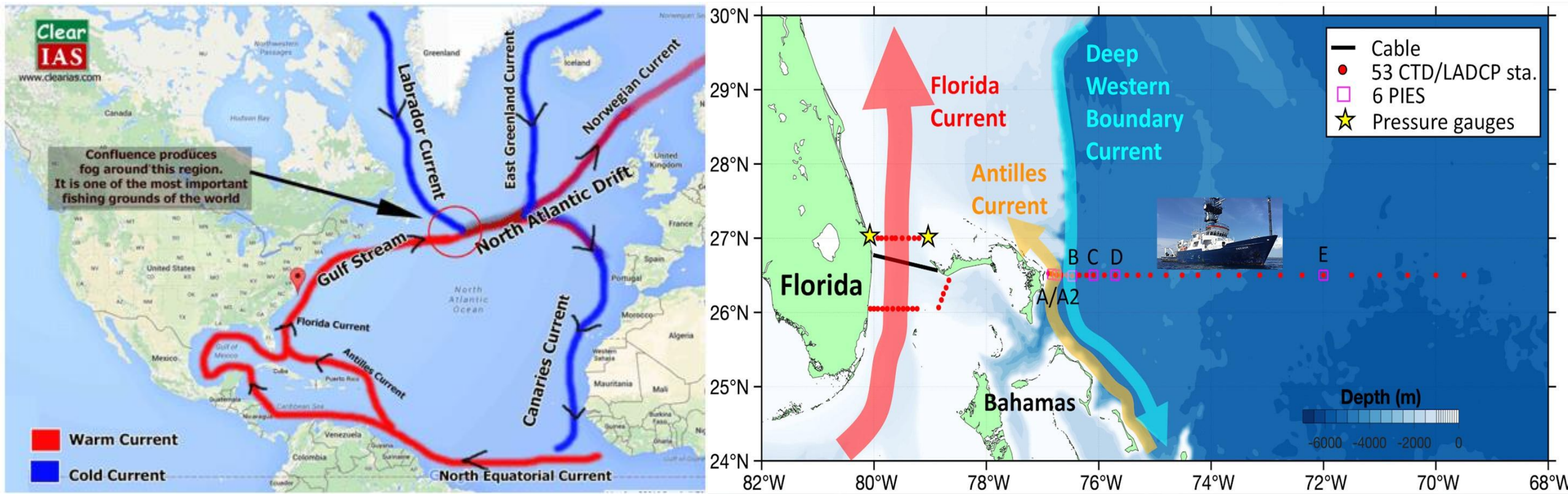
- The Northeast Monsoon Current is westward only during the months of January through March, and is strongest in February
- The monsoon currents are the seasonally reversing, open-ocean currents that flow between the Arabian Sea and the Bay of Bengal, the two wings of the north Indian Ocean. The Summer Monsoon Current (SMC) flows eastward during the summer monsoon (May–September) and the Winter Monsoon Current (WMC) flows westward during the winter monsoon (November–February).





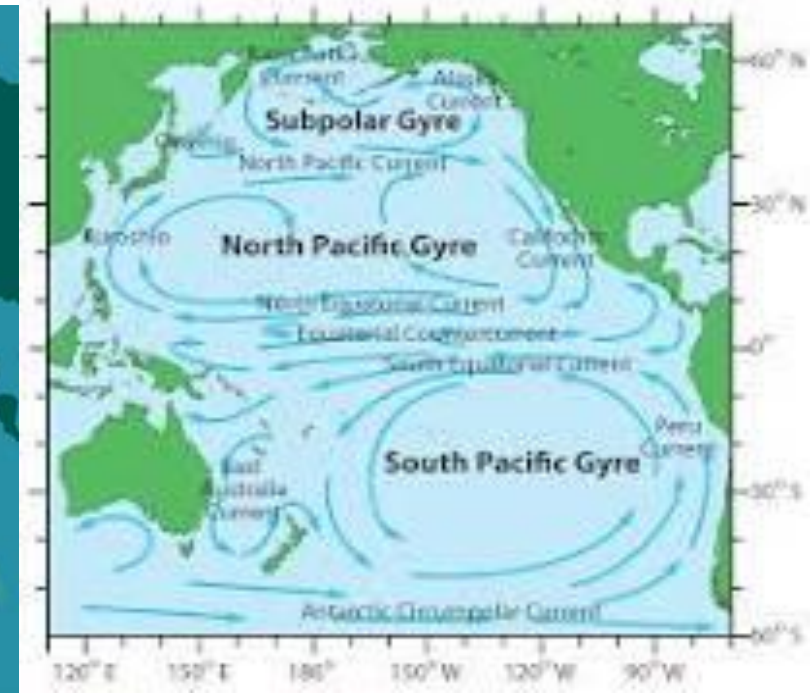
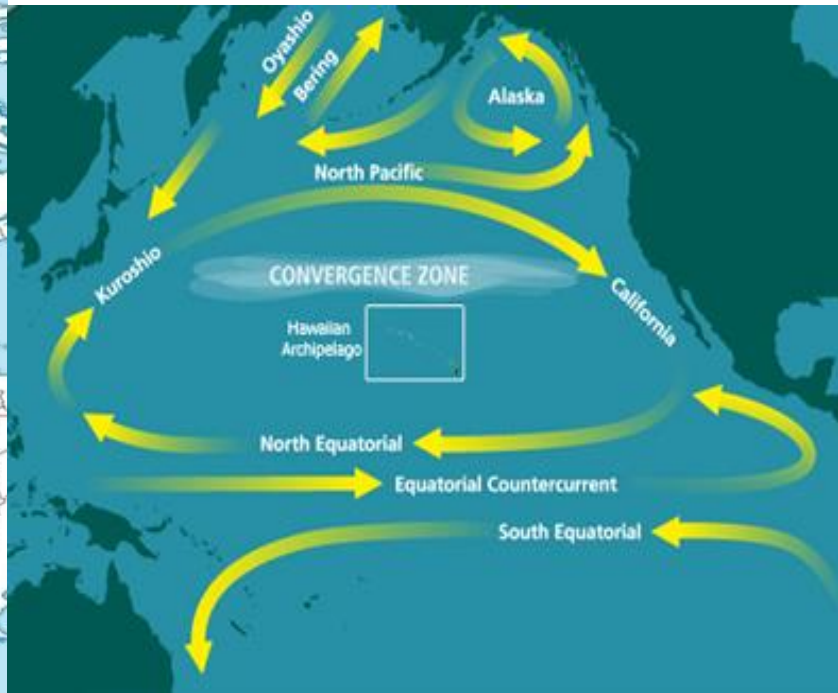
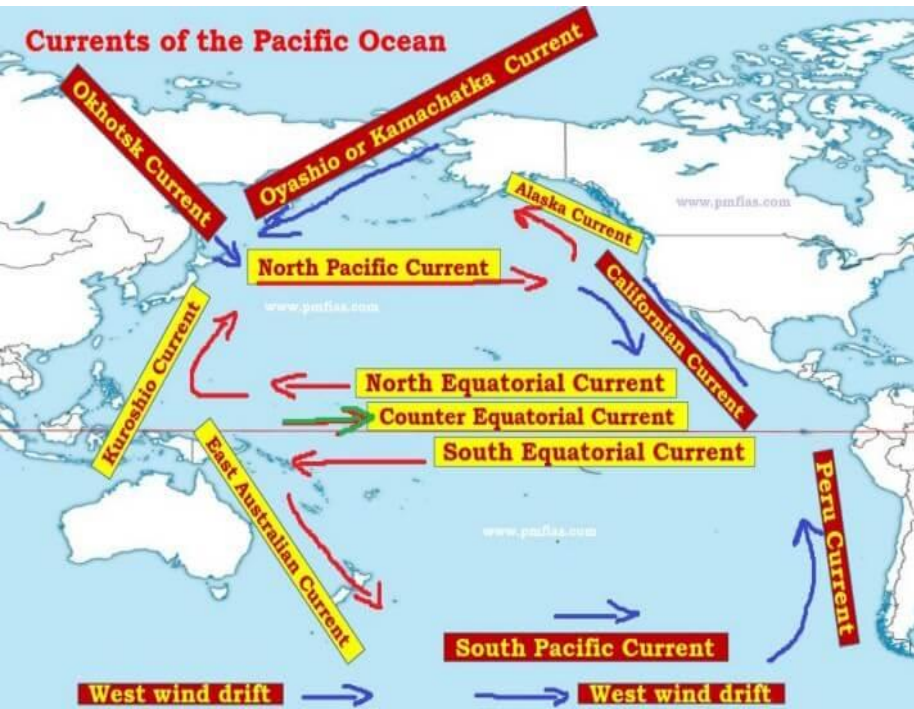
FLORIDA WARM CURRENT

- The Florida Current is a strong oceanic current flowing northward along the eastern coast of Florida carrying warm tropical waters that eventually feed the Gulf Stream
- If the Florida Current is slowing down, it could mean the Atlantic's larger circulation system is weakening. That could cause floods, droughts and sea rise among other impacts



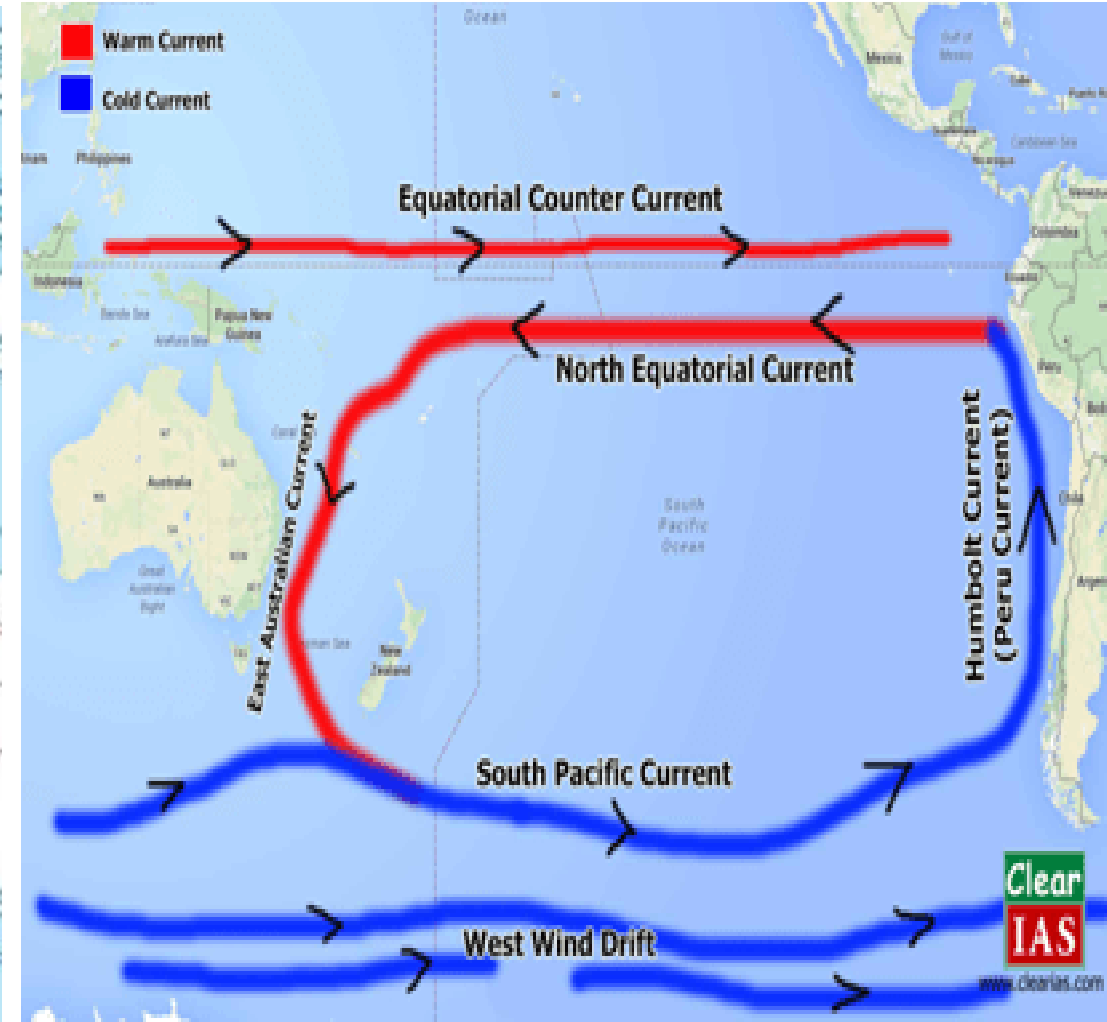
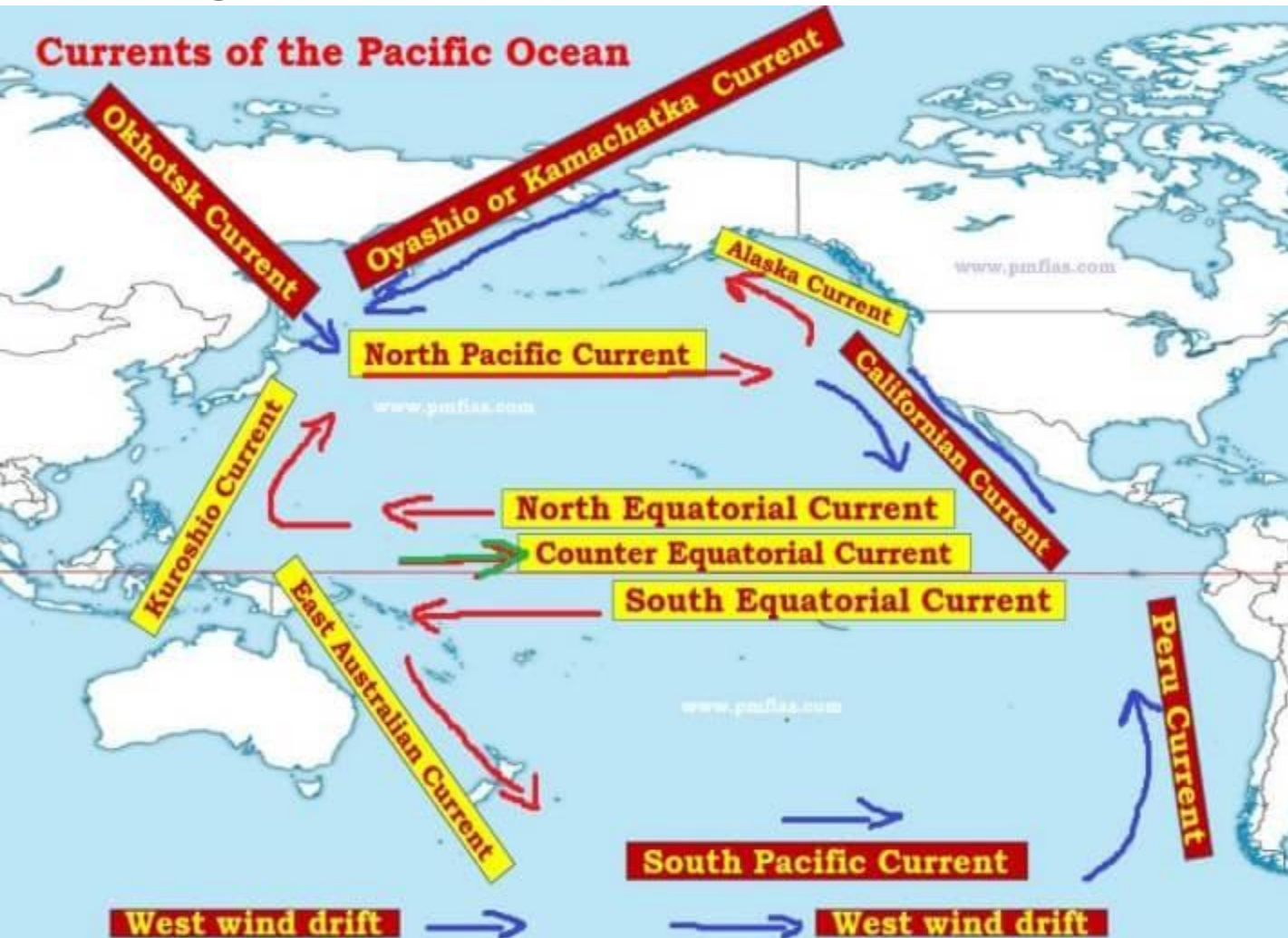
North Pacific Current

- It is formed by the collision of Kuroshio & Oyashio. It circulates counterclockwise along the Western North Pacific Ocean.
- Flowing in an eastern direction along the northern periphery of the Pacific Ocean, the North Pacific Current carries moderately warm water from the western Pacific toward the shores of North America
- The North Pacific Current (sometimes referred to as the North Pacific Drift) is an ocean current that flows west-to-east between 30 and 50 degrees north in the Pacific Ocean



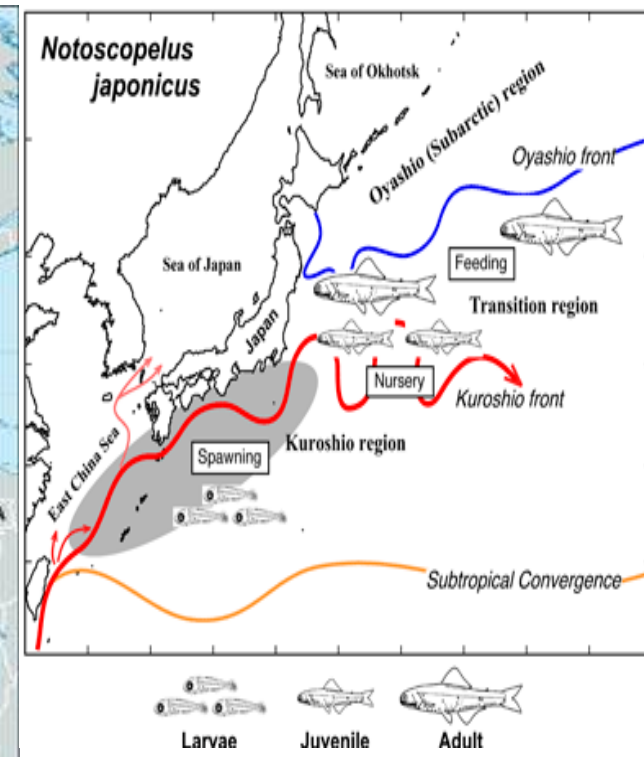
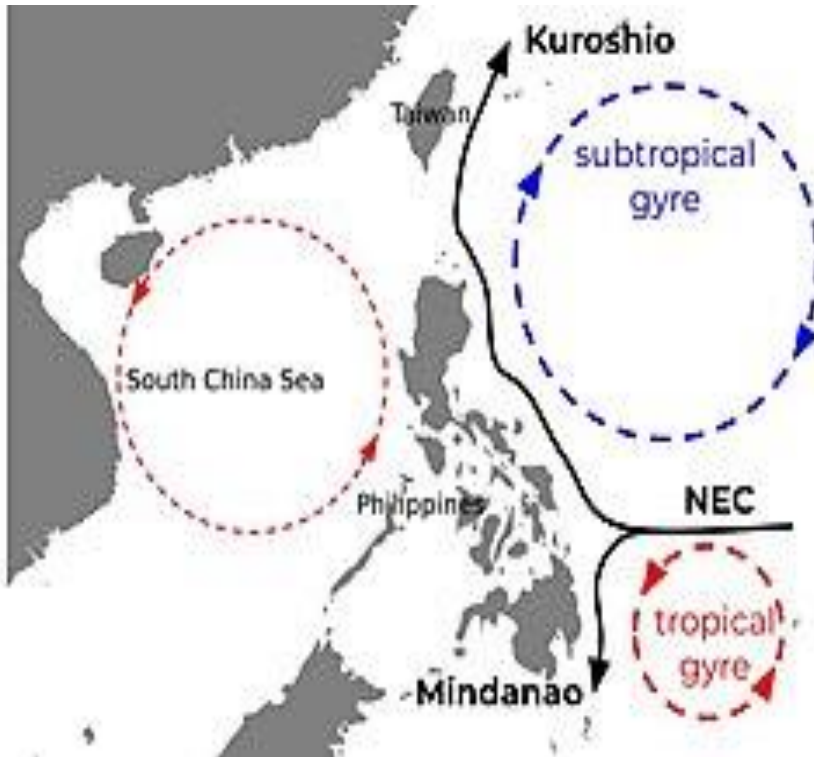
NORTH EQUATORIAL WARM CURRENT

- The water of this current is relatively warm as compared to the surrounding waters in this zone. The southern branch of the current moves as a cold current along the west coast of USA and is known as the Californian current.



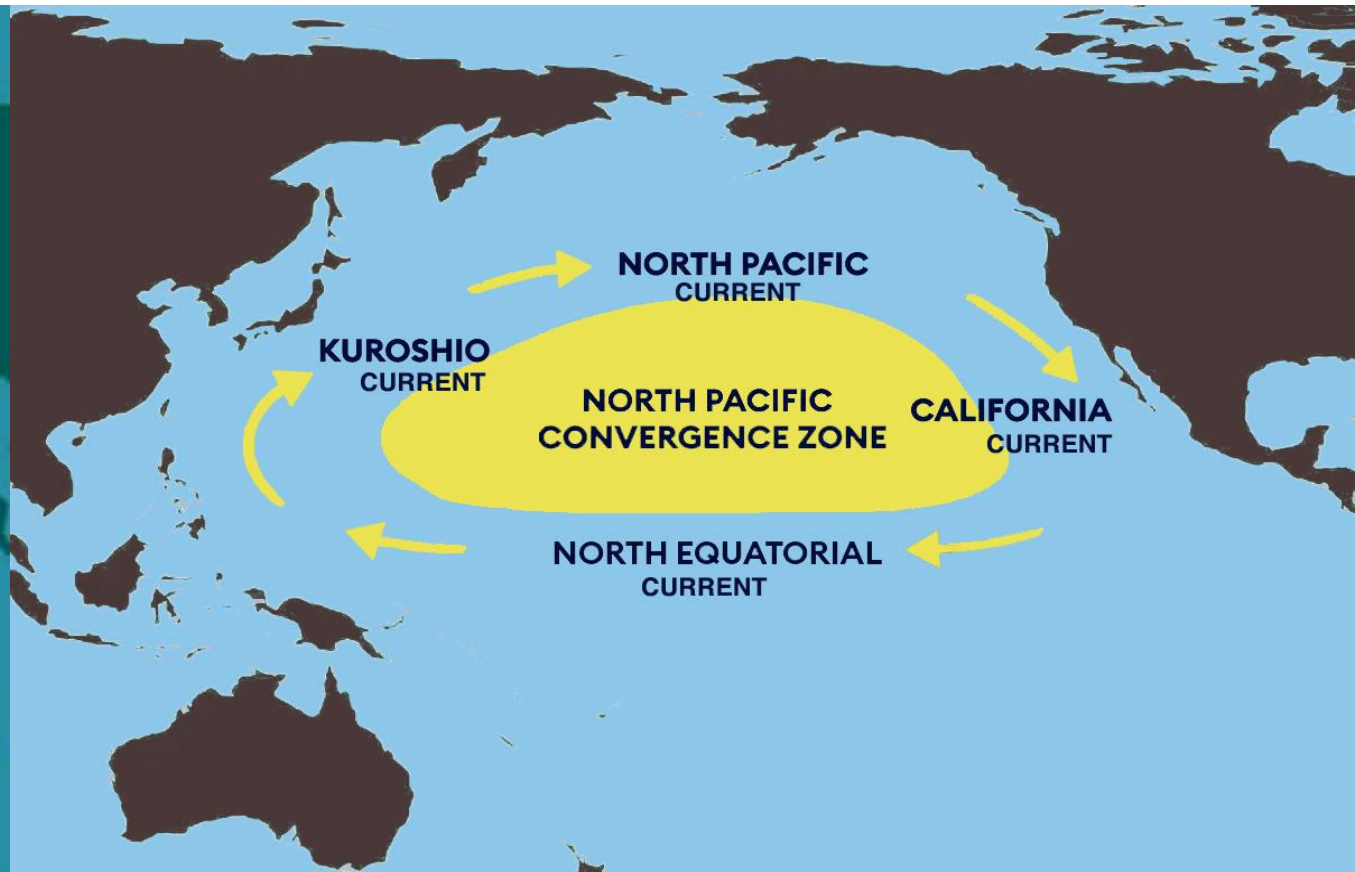
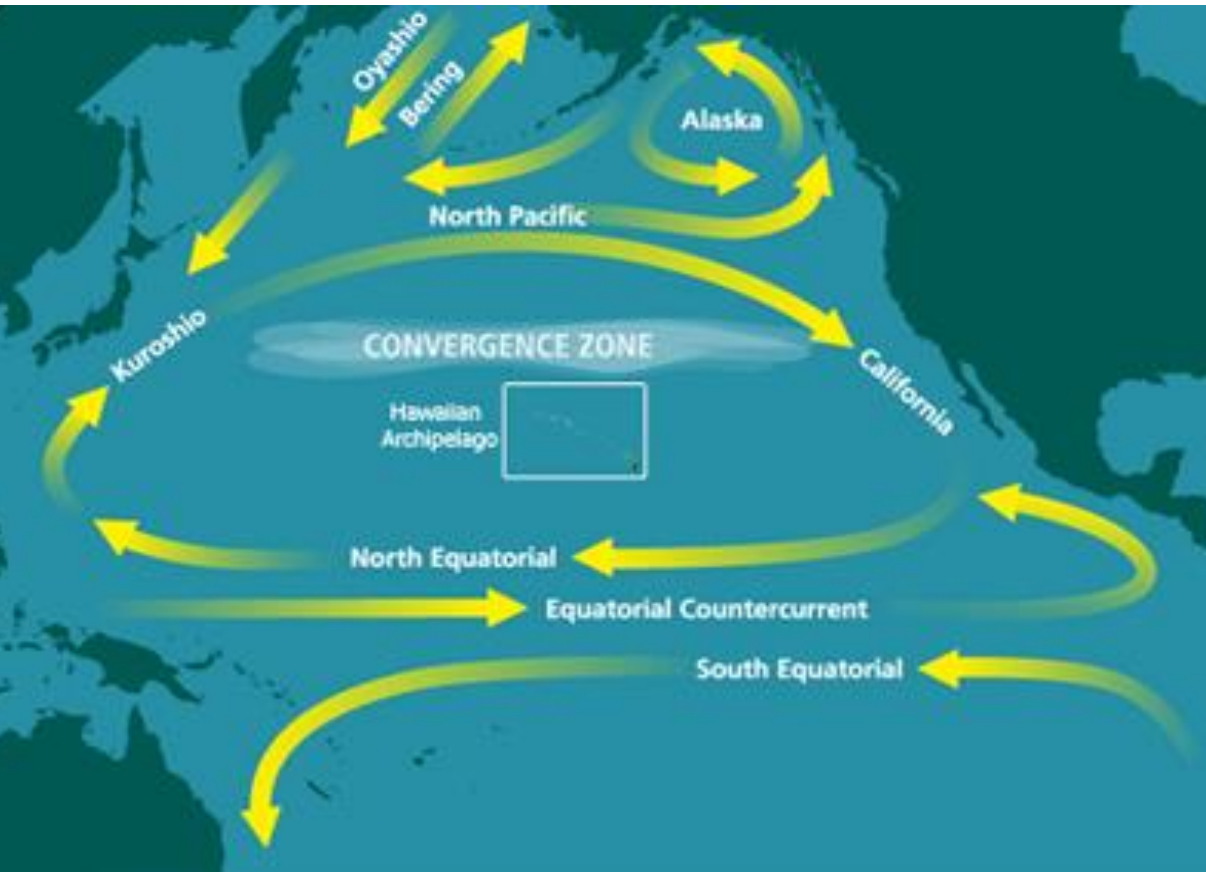
KURISHIO WARM CURRENT

- The Kuroshio is a relatively warm ocean current with an annual average sea-surface temperature of about 24 °C (75 °F), is approximately 100 kilometres (62 mi) wide, and produces frequent small to meso-scale eddies.
- The Kuroshio Current is one of the primary large-scale currents that affect the Northwestern Hawaiian Islands



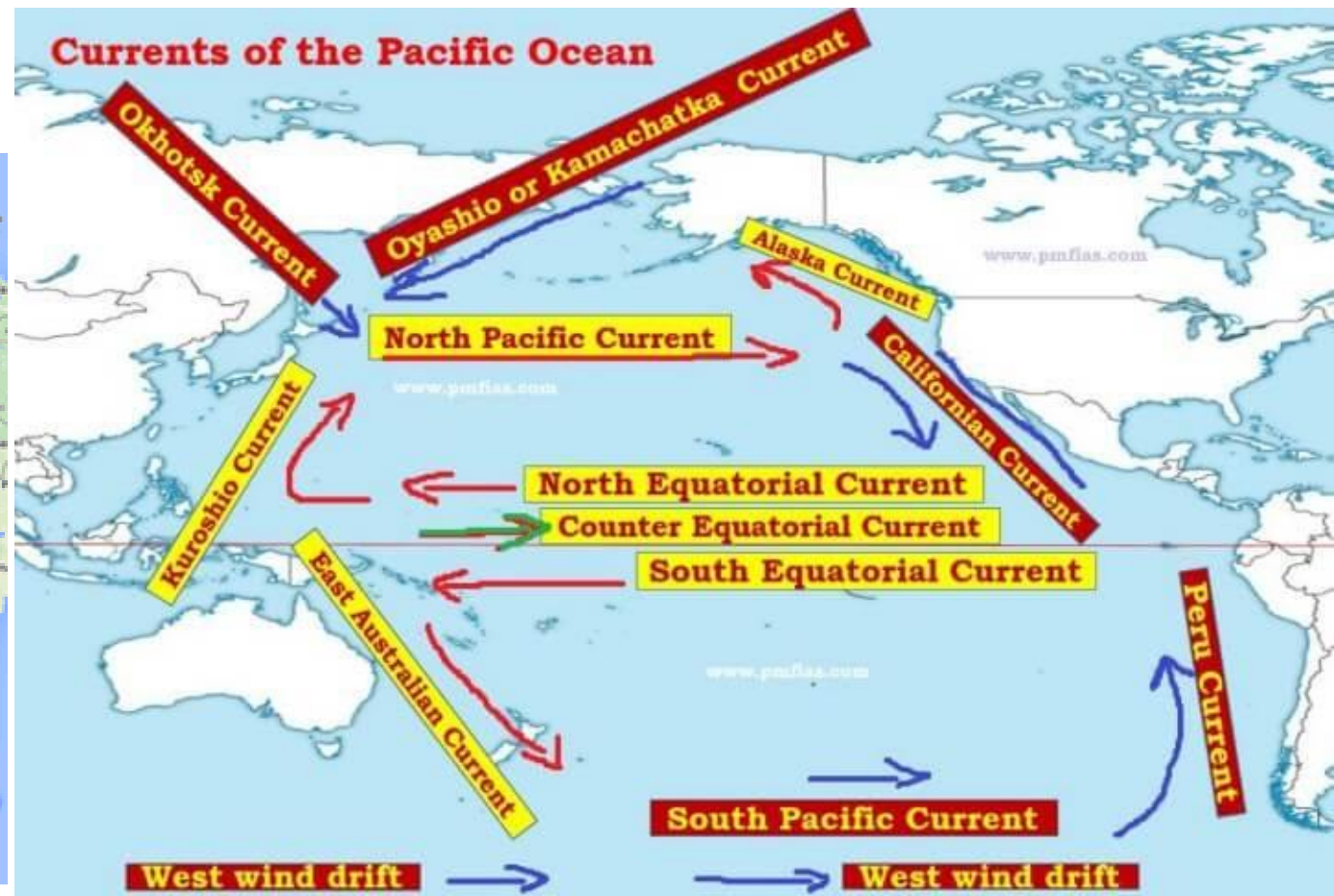
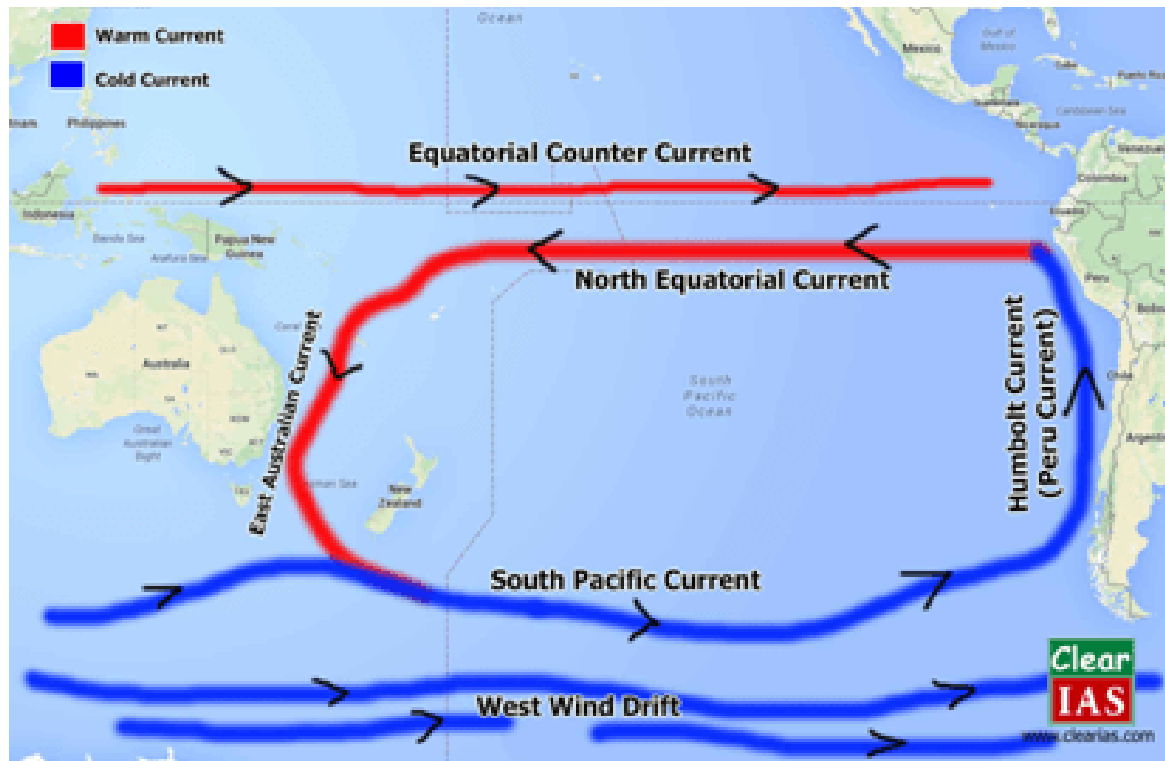
NORTH PACIFIC WARM CURRENT

- Flowing in an eastern direction along the northern periphery of the Pacific Ocean, the North Pacific Current carries moderately warm water from the western Pacific toward the shores of North America.
- Its influence contributes to the mild climate along the western coast of the United States and Canada.



COUNTER EQUATORIAL WARM CURRENT

- The Equatorial Counter Current is a weaker, eastward-flowing current located between the powerful westward-flowing North and South Equatorial Currents. It flows eastward because the westward trade winds pile up a large volume of warm water in the western parts of ocean basins.

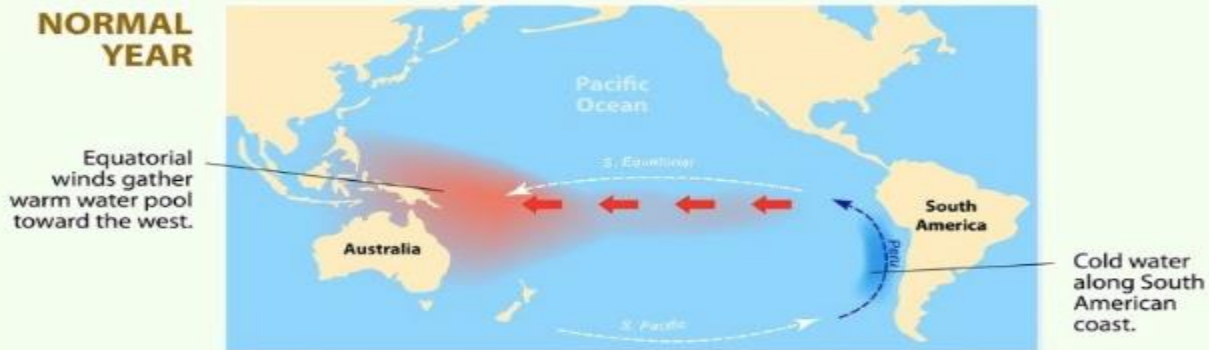


EL NINO WARM CURRENT

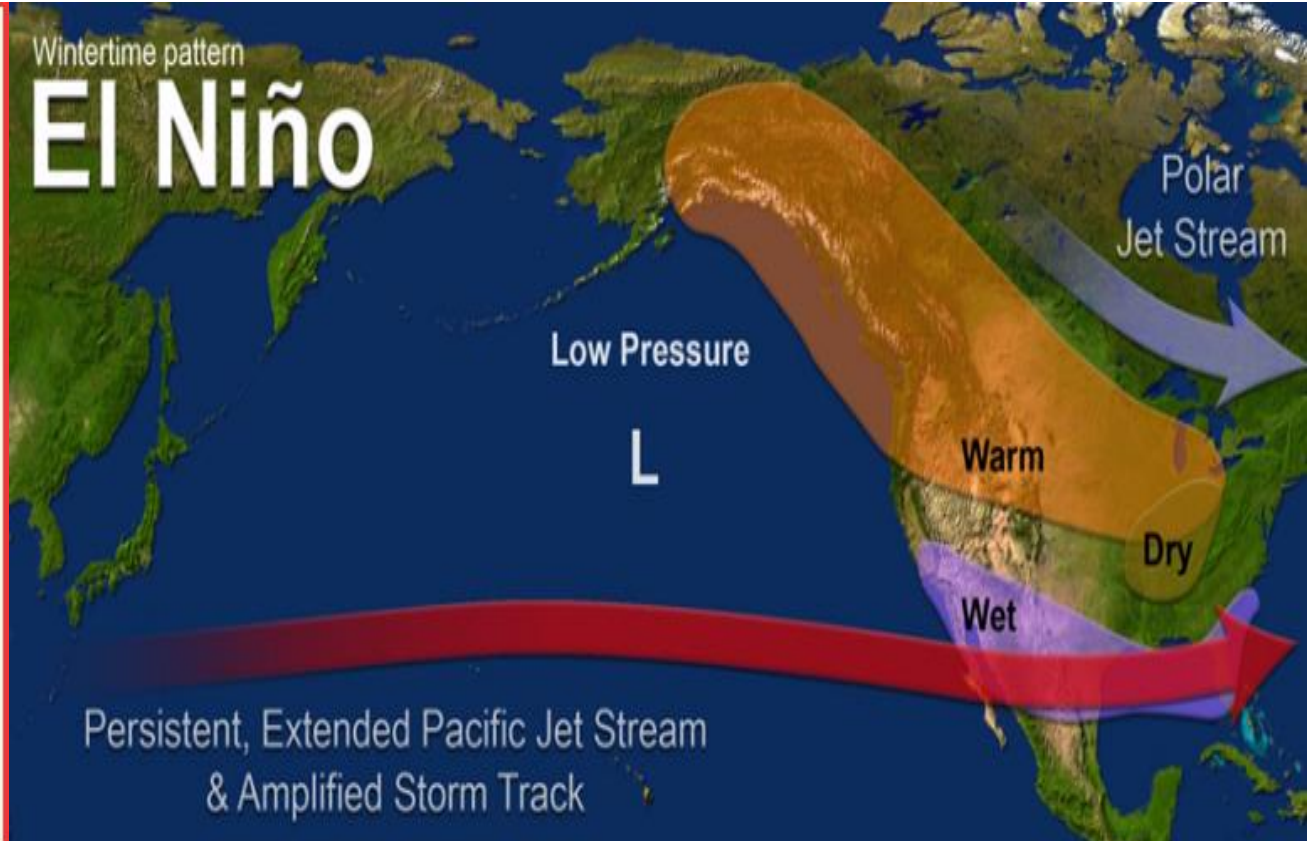
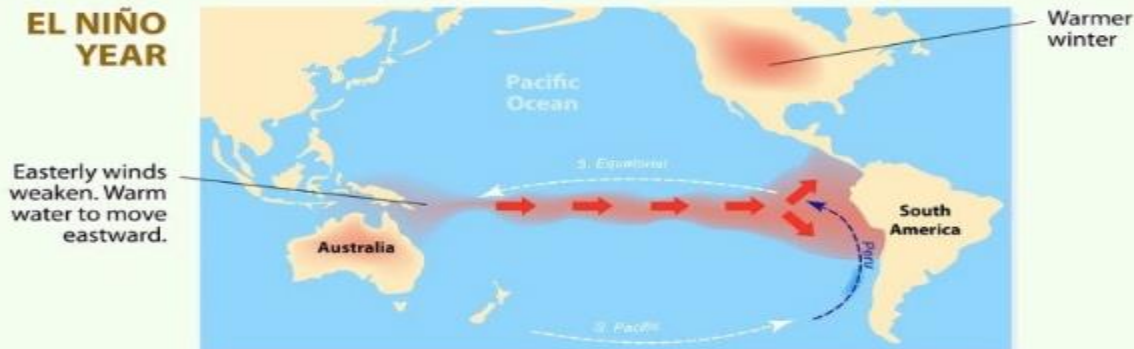
- Actually El Nino is a term used to describe the warm southward current of the Pacific Ocean, that appears every December of the year.
- El Nino, meaning "Child Christ" is a warm ocean current appearing along the Peru coast, generally in December. It normally happens once every 4-5 years though it is not fixed.

THE EL NIÑO PHENOMENON

NORMAL YEAR



EL NIÑO YEAR



AGULHAS WARM CURRENT

- The Agulhas Current is a **warm, fast-flowing western boundary current** in the southwestern Indian Ocean. It flows southward along the east coast of Africa, transporting warm, subtropical water from the Mozambique Channel and east of Madagascar toward the southern tip of Africa.

